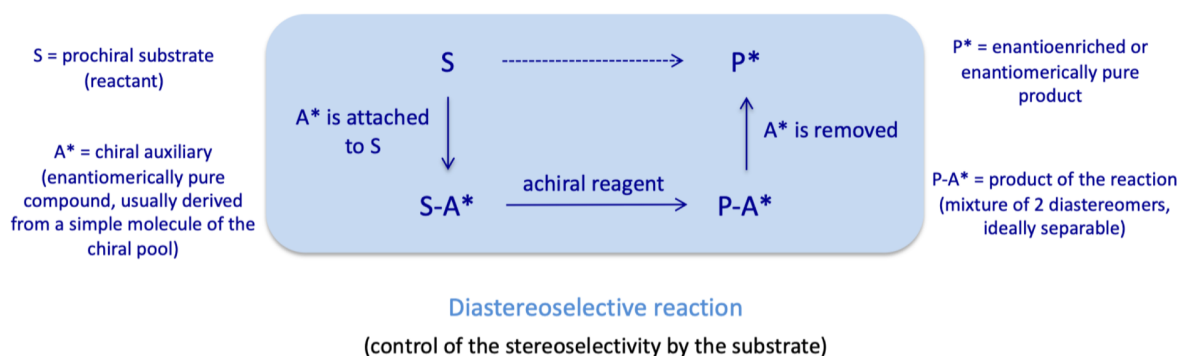


Chap 4: Asymmetric synthesis: An introduction

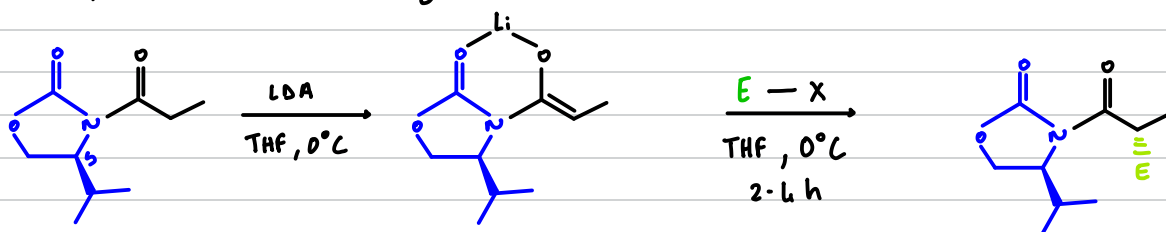
In asymmetric synthesis, stereoselectivity is mainly influenced by the chiral auxiliary, the chiral reagent or the chiral catalyst.

1. The chiral auxiliary strategy



Advantages: the control of the diastereoselectivity is generally good - the obtained diastereomers can often be separated.

Disadvantages: multi-sep reactⁿ · stoichiometric reaction - need to have access to both enantiomers of the chiral auxiliary.



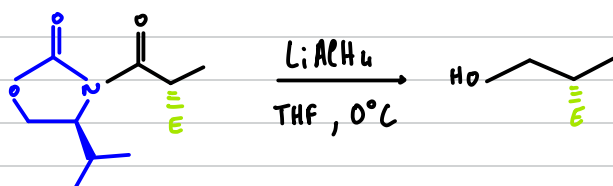
Evans chiral auxiliary

Cis enolate
(Z enolate)
on obtient
uniquement l'énolate cis
c'est le model de Halland

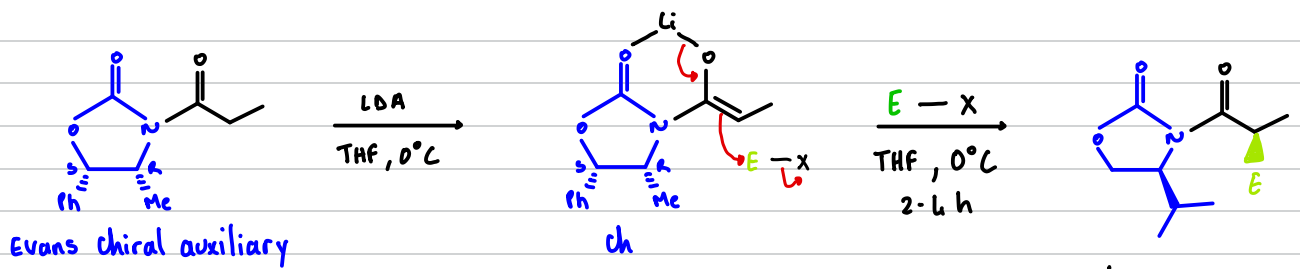
Major diastereomer

E = PhCH ₂	d.r. : 99:1 92%
CH ₂ =CH=CH ₂	d.r. : 98:2 74%
Et	d.r. : 94:6 36%

isolated yield for the major diastereomer



- ici on a obtenu un alcool, mais si on voulait un aldéhyde il fallait utiliser du DIBAL, pour avoir un ester on aurait utilisé un alcoolate.



Major diastereomer

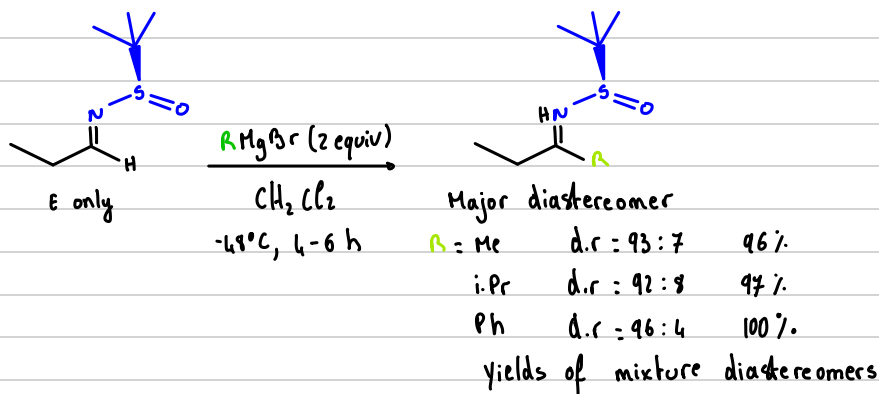
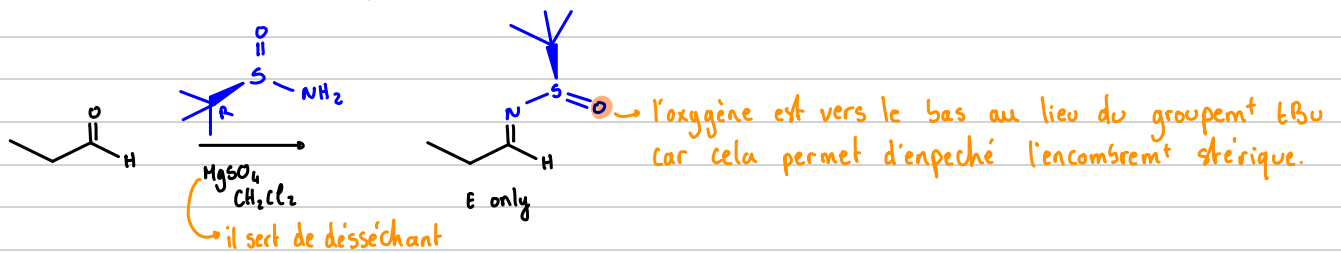
E = PhCH₂

CH₂=CH=CH₂

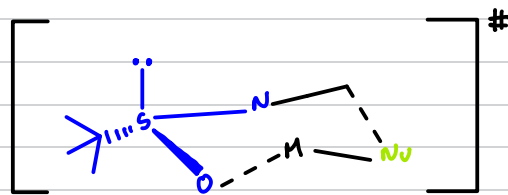
Et

isolated yield for the major diastereomer

Ellman chiral auxiliary



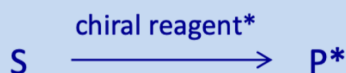
Zimmerman-Traxler
Chelated transit^o-state



Addition on the Si face

2. The chiral reagent strategy

S = prochiral substrate
(reactant)



P* = enantioenriched
product

chiral reagent* =
enantiopure reagent,
used in stoichiometric amount

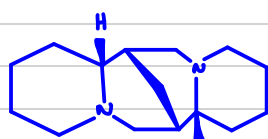
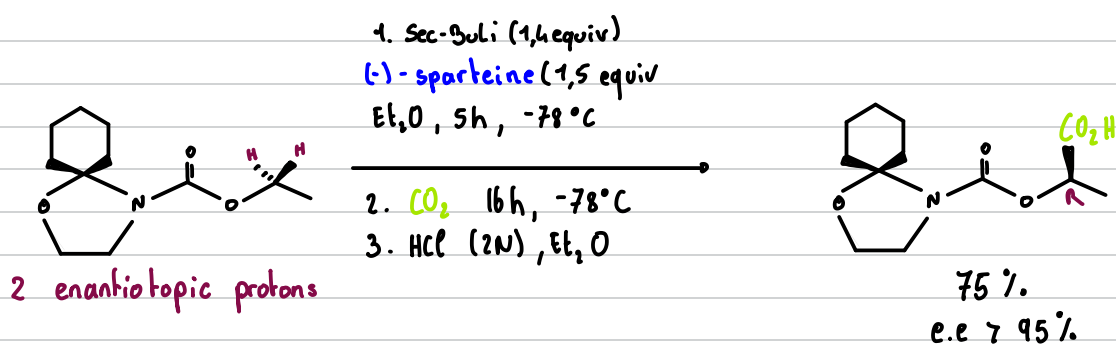
Enantioselective reaction

(control of the stereoselectivity by the reagent)

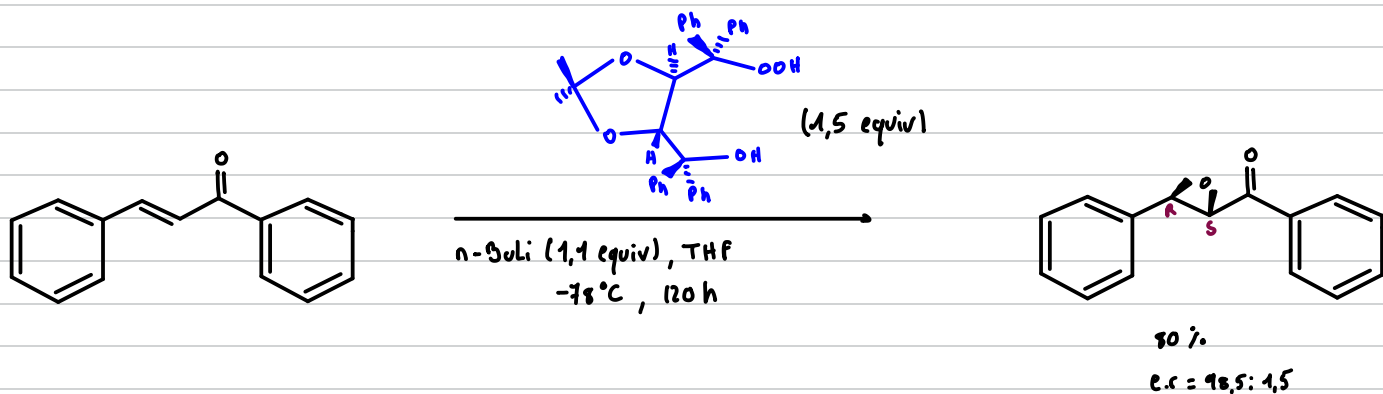
Advantages: One step reaction

↳ greener approach than the chiral auxiliary approach

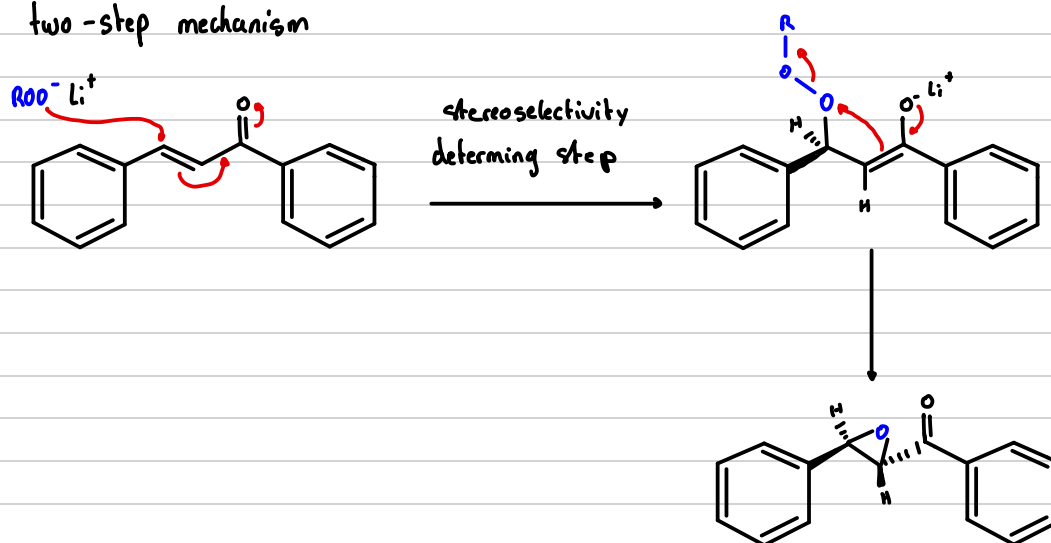
Disadvantages: stoichiometric amount of reagent required - in most cases the reagent need to be prepared - product obtained as mixture of enantiomers which could not be separated easily.



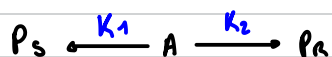
(-)-sparteine



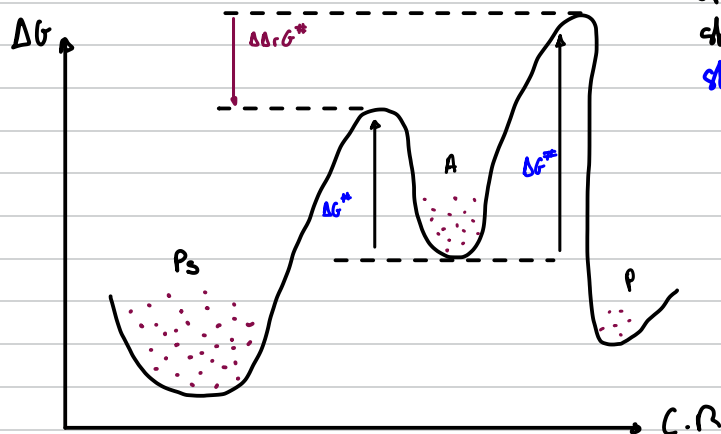
two-step mechanism



Enantioselective reactions are under kinetic control



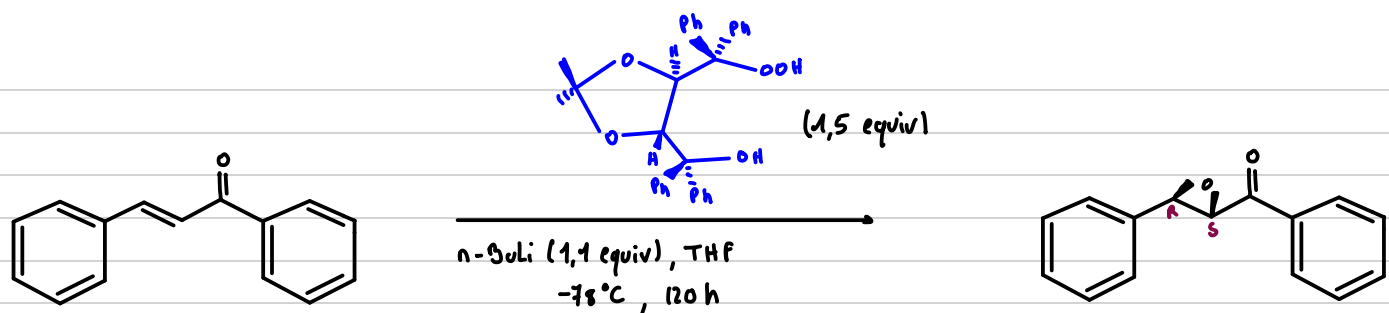
One step mechanism



the use of chiral reagent favors the format of one enantiomer over the other one through interact at the transition state (the TS must be necessarily diastereoisomeric $\Delta\Delta G^\ddagger \neq 0$)

$$\frac{[P_S]}{[P_R]} = \frac{k_1}{k_2} = e^{-\frac{\Delta\Delta G^\ddagger}{RT}}$$

the temperature of the reaction has an influence on the enantiomeric ratio.



0°C 6h 96% e.r. = 90:10
 -30°C 24h 92% e.r. = 95:5
 -78°C 120h 80% e.r. = 98,5:1,5

Quand on effectue des réact° énantiométriques on ne chauffe pas.

3. The chiral catalyst strategy

S = prochiral substrate (reactant)



P^* = enantioenriched product

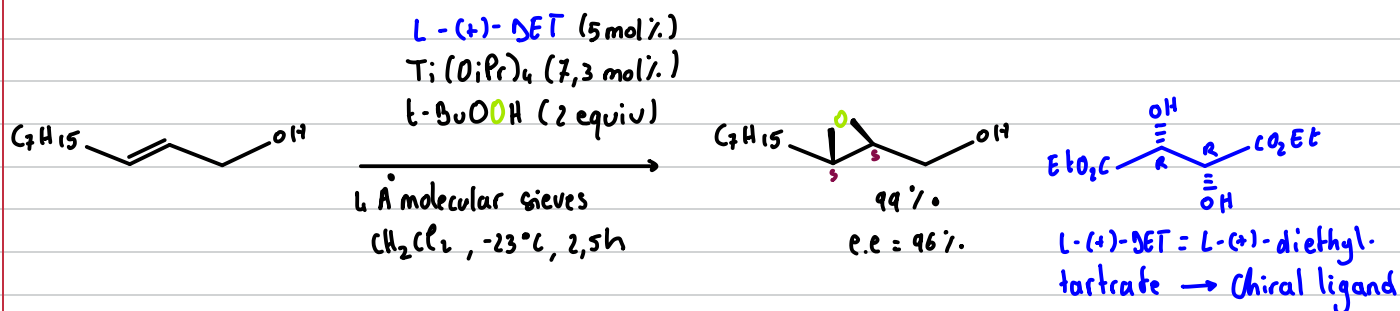
chiral catalyst* =
 enantiopure reagent or ligand,
 used in catalytic amount (at the end of the reaction
 the catalyst should return to its initial state)

Enantioselective reaction

(control of the stereoselectivity by the catalyst)

Advantages: Very elegant method - one step reaction - use of chiral reagent in very low loading.
 → greenest strategy (catalysis is one of the 12 principles of green chemistry)

Disadvantages: highly specific catalyst - identification of the best catalyst is time-consuming - catalysts are often expensive - product obtained as mixture of enantiomers which could not be separated easily.



transition metal-catalysed pathway (chapter 2).